

Navigators Research Paper

AI-Assisted Dead Reckoning, Non-Holonomic Kinematics & Adaptive Inertial Sensor Fusion

Navigators : Navigation Mathematics & Theoretical Formulation

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1. Mathematical Notation & Coordinate Frames

1.1 Notation Glossary

- Lowercase boldface ($\mathbf{x}$, $\mathbf{p}$, $\mathbf{v}$): Vectors in $\mathbb{R}^n$.
- Uppercase boldface ($\mathbf{R}$, $\mathbf{P}$, $\mathbf{F}$, $\mathbf{H}$): Matrices in $\mathbb{R}^{m \times n}$.
- Subscripts (k , $k-1$): Discrete temporal epochs.
- Superscripts ($\{b\}$, $\{v\}$, $\{n\}$, $\{e\}$): Coordinate frames of reference.
- $\mathbf{R}_a^b$: Direction Cosine Matrix (DCM) transforming a vector from frame $\{a\}$ to frame $\{b\}$.
- $\mathbf{v}_\times$: Skew-symmetric cross-product matrix for $\mathbf{v} = [v_1, v_2, v_3]^T$:
$$\mathbf{v}_\times = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}$$

1.2 Coordinate Reference Frames

- Device Body Frame ($\{b\}$)**: Orthogonal sensor triad aligned with physical smartphone casing.
- Vehicle Body Frame ($\{v\}$)**: Orthogonal axes aligned with vehicle dynamics (X_v : transverse/right, Y_v : forward/longitudinal, Z_v : vertical/up).
- Local Tangent Frame ($\{n\}$ - ENU)**: Local Cartesian East-North-Up frame tangent to Earth reference ellipsoid at initial GNSS anchor (ϕ_0, λ_0, h_0) .
- Earth Frame ($\{e\}$ - WGS-84)**: Global geocentric reference frame defined by semi-major axis $a = 6378137.0$ (m) and flattening $f = 1 / 298.257223563$.

2. Phone-to-Vehicle Alignment Mathematics

Let $\mathbf{f}^b$ denote the specific force measured by the tri-axial accelerometer in the device frame, and $\boldsymbol{\omega}^b$ denote the angular velocity measured by the gyroscope.

2.1 Gravity Vector Extraction (Stationary Dwell)

During zero-velocity conditions prior to vehicle movement:

$$\mathbf{f}_{\text{dwell}}^b = -\mathbf{R}_n^b \mathbf{g}^n + \mathbf{b}_a + \mathbf{w}_a$$

Averaging across N_{cal} dwell samples yields estimated downward gravity unit vector:

$$\mathbf{u}_{\text{down}}^b = \frac{1}{N_{\text{cal}}} \sum_{i=1}^{N_{\text{cal}}} \mathbf{f}_i^b \quad \left| \frac{1}{N_{\text{cal}}} \sum_{i=1}^{N_{\text{cal}}} \mathbf{f}_i^b \right|$$

2.2 Forward Acceleration Vector Extraction

When the vehicle accelerates forward along longitudinal axis Y_v :

$$\mathbf{a}_{\text{net}}^b = \mathbf{f}_{\text{motion}}^b - \mathbf{g}^b \mathbf{u}_{\text{down}}^b$$

Projecting orthogonal to the downward vector:

$$\mathbf{u}_{\text{fwd}}^b = \frac{\mathbf{a}_{\text{net}}^b - (\mathbf{a}_{\text{net}}^b \cdot \mathbf{u}_{\text{down}}^b) \mathbf{u}_{\text{down}}^b}{\|\mathbf{a}_{\text{net}}^b - (\mathbf{a}_{\text{net}}^b \cdot \mathbf{u}_{\text{down}}^b) \mathbf{u}_{\text{down}}^b\|}$$

2.3 Triad Construction of Rotation Matrix $\mathbf{R}_b^v$

$$\mathbf{r}_1 = \mathbf{u}_{\text{fwd}}^b \quad (\text{Longitudinal / Forward})$$

$$\mathbf{r}_2 = \frac{\mathbf{u}_{\text{fwd}}^b \times \mathbf{u}_{\text{down}}^b}{\|\mathbf{u}_{\text{fwd}}^b \times \mathbf{u}_{\text{down}}^b\|} \quad (\text{Transverse / Right})$$

$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2 \quad (\text{Vertical / Up})$$

The complete orthogonal transformation from phone body to vehicle body is:

$$\mathbf{R}_b^v = \begin{bmatrix} \mathbf{r}_2^T \\ \mathbf{r}_1^T \\ \mathbf{r}_3^T \end{bmatrix} \in \text{SO}(3)$$

3. Kinematic Navigation State Equations

3.1 State Vector Formulation

The continuous 15-state vector $\mathbf{x}(t)$ is formulated as:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{p}^n(t) \\ \mathbf{v}^n(t) \\ \boldsymbol{\theta}(t) \\ \mathbf{b}_a(t) \\ \mathbf{b}_g(t) \end{bmatrix} = \begin{bmatrix} p_E, p_N, p_U \\ v_E, v_N, v_U \\ \phi, \theta, \psi \\ b_{ax}, b_{ay}, b_{az} \\ b_{gx}, b_{gy}, b_{gz} \end{bmatrix} \in \mathbb{R}^{15}$$

3.2 Continuous-Time Differential Equations

$$\dot{\mathbf{p}}^n(t) = \mathbf{v}^n(t)$$

$$\dot{\mathbf{v}}^n(t) = \mathbf{R}_b^n(t) (\mathbf{f}^b(t) - \mathbf{b}_a(t) - \mathbf{w}_a(t)) + \mathbf{g}^n(t)$$

$$\dot{\boldsymbol{\theta}}(t) = \mathbf{E}(\boldsymbol{\theta}(t))(\boldsymbol{\omega}^b(t) - \mathbf{b}_g(t) - \mathbf{w}_g(t))$$

$$\dot{\mathbf{b}}_a(t) = \mathbf{w}_{ba}(t)$$

$$\dot{\mathbf{b}}_g(t) = \mathbf{w}_{bg}(t)$$

where $\mathbf{g}^n = [0, 0, -9.81]^T \text{ m/s}^2$, and $\mathbf{E}(\boldsymbol{\theta})$ maps body angular rates to Euler angle rates:

$$\mathbf{E}(\phi, \theta, \psi) = \begin{bmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} & \sec\theta \end{bmatrix}$$

The rotation matrix $\mathbf{R}_b^n$ as a function of Euler angles (ϕ, θ, ψ) follows the standard Z-Y-X convention:

$$\mathbf{R}_b^n = \begin{bmatrix} \cos\psi \cos\theta & \cos\psi \sin\theta \sin\phi & \sin\psi \cos\phi \\ \sin\psi \cos\theta & \sin\psi \sin\theta \sin\phi & \cos\psi \cos\phi \\ -\sin\theta & \cos\theta \sin\phi & \cos\theta \cos\phi \end{bmatrix}$$

4. Discrete Extended Kalman Filter Formulation

4.1 State Transition Matrix $\mathbf{F}_k$

Linearizing the continuous error dynamics over timestep Δt :

$$\mathbf{F}_k = \mathbf{I}_{15} + \mathbf{A}_k \Delta t$$

where the non-zero sub-blocks of Jacobian $\mathbf{A}_k$ are:

$$\mathbf{A}_{0:3, 3:6} = \mathbf{I}_{3 \times 3}$$

$$\mathbf{A}_{\{3:6, 6:9\}} = -\mathbf{R}_b^n \hat{\mathbf{f}}^b \times$$

$$\mathbf{A}_{\{3:6, 9:12\}} = -\mathbf{R}_b^n$$

$$\mathbf{A}_{\{6:9, 6:9\}} \approx \mathbf{0}_{\{3 \times 3\}} \quad (\text{small angle approximation})$$

$$\mathbf{A}_{\{6:9, 12:15\}} = -\mathbf{E}(\hat{\boldsymbol{\theta}})$$

4.2 Discrete Process Noise Covariance $\mathbf{Q}_k$

$$\mathbf{Q}_k = \text{diag}(\mathbf{q}_{\text{pos}} \Delta t^2 \mathbf{I}_3, \mathbf{q}_{\text{vel}} \Delta t \mathbf{I}_3, \mathbf{q}_{\text{ori}} \Delta t \mathbf{I}_3, \mathbf{q}_{\text{ba}} \Delta t \mathbf{I}_3, \mathbf{q}_{\text{bg}} \Delta t \mathbf{I}_3)$$

Default spectral densities in the repository:

- $\mathbf{q}_{\text{pos}} = 0.5 \text{ m}^2/\text{s}^4$
- $\mathbf{q}_{\text{vel}} = 2.0 \text{ m}^2/\text{s}^3$
- $\mathbf{q}_{\text{ori}} = 0.05 \text{ rad}^2/\text{s}$
- $\mathbf{q}_{\text{ba}} = 10^{-3} \text{ m}^2/\text{s}^5$
- $\mathbf{q}_{\text{bg}} = 10^{-4} \text{ rad}^2/\text{s}^3$

4.3 State & Covariance Time Propagation

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{x}}_{k-1}, \mathbf{f}_k^b, \boldsymbol{\omega}_k^b, \Delta t)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q}_k$$

5. Multi-Sensor Measurement Updates

When an observation $\mathbf{z}_k$ occurs with measurement model $\mathbf{h}(\mathbf{x}_k)$:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}) \quad (\text{Innovation})$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k \quad (\text{Innovation Covariance})$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad (\text{Kalman Gain})$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k$$

5.1 Joseph-Form Covariance Update

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^T$$

This symmetric formulation guarantees positive semi-definiteness regardless of rounding errors in single-precision floating point.

5.2 GNSS Fusion Update

When satellite reception is active:

$$\mathbf{z}_{\text{gnss}} = [\mathbf{p}_E^{\text{gps}}, \mathbf{p}_N^{\text{gps}}, \mathbf{p}_U^{\text{gps}}, \mathbf{v}_E^{\text{gps}}, \mathbf{v}_N^{\text{gps}}, \mathbf{v}_U^{\text{gps}}]^T \in \mathbb{R}^6$$

$$\mathbf{H}_{\text{gnss}} = \begin{bmatrix} \mathbf{I}_{\{3 \times 3\}} & \mathbf{0}_{\{3 \times 3\}} & \mathbf{0}_{\{3 \times 9\}} \\ \mathbf{0}_{\{3 \times 3\}} & \mathbf{I}_{\{3 \times 3\}} & \mathbf{0}_{\{3 \times 9\}} \end{bmatrix} \in \mathbb{R}^{6 \times 15}$$

$$\mathbf{R}_{\text{gnss}} = \text{diag}(\sigma_{\text{pos}}^2 \mathbf{I}_3, \sigma_{\text{vel}}^2 \mathbf{I}_3)$$

5.3 AI Velocity Pseudo-Measurement Update

During GNSS denial, the TCN model outputs 2D planar velocity $[\mathbf{v}_N^{\text{tcn}}, \mathbf{v}_E^{\text{tcn}}]$:

$$\mathbf{z}_{\text{ai}} = [\mathbf{v}_E^{\text{tcn}}, \mathbf{v}_N^{\text{tcn}}]^T \in \mathbb{R}^2$$

$$\mathbf{H}_{\text{ai}} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & \mathbf{0}_{\{1 \times 9\}} \\ \mathbf{0}_{\{1 \times 9\}} \end{bmatrix} \in \mathbb{R}^{2 \times 15}$$

$$\mathbf{R}_{\text{ai}} = \text{diag}(\sigma_{\text{ai}}^2, \sigma_{\text{ai}}^2), \quad \sigma_{\text{ai}} = 0.3 \text{ m/s}$$

5.4 Non-Holonomic Constraints (NHC)

Assuming land vehicle non-slip dynamics:

$$v_{\text{lateral}} = -v_E \sin\psi + v_N \cos\psi \approx 0$$

$$v_{\text{vertical}} = v_U \approx 0$$

$$\mathbf{z}_{\text{nhc}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{H}_{\text{nhc}} = \begin{bmatrix} 0 & 0 & 0 & -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_{\text{nhc}} = \text{diag}(1.0, 1.0) \text{ m}^2/\text{s}^2$$

5.5 Zero-Velocity Updates (ZUPT)

During stationary dwell:

$$\mathbf{z}_{\text{zupt}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{H}_{\text{zupt}} = [\mathbf{0}_{3 \times 3}, \mathbf{0}_{3 \times 9}]$$

$$\mathbf{R}_{\text{zupt}} = \text{diag}(0.01, 0.01, 0.01) \text{ m}^2/\text{s}^2$$

6. Geodetic Coordinate Transformations

6.1 WGS-84 to Local ENU

Given geodetic coordinate (ϕ, λ, h) and reference origin (ϕ_0, λ_0, h_0) :

- Difference in radians: $\Delta\phi = \phi - \phi_0$, $\Delta\lambda = \lambda - \lambda_0$
- Meridional radius of curvature M and prime vertical radius N :

$$M = \frac{a(1 - e^2)}{(1 - e^2 \sin^2\phi_0)^{3/2}}, \quad N = \frac{a}{\sqrt{1 - e^2 \sin^2\phi_0}}$$

where eccentricity squared $e^2 = 2f - f^2$.

- Local Cartesian ENU coordinates:

$$p_E = (N + h_0) \cos\phi_0 \Delta\lambda$$

$$p_N = (M + h_0) \Delta\phi$$

$$p_U = h - h_0$$

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